

Senior Data Engineer Interview Preparation Guide

Technical: Scalable Data Pipelines

- How do you design a data pipeline to handle both structured and unstructured data?
- What are the key considerations for building scalable and fault-tolerant data pipelines?
- Can you describe a pipeline you built that ingested data from multiple sources? What challenges did you face?

Technical: PySpark & Spark SQL in Azure Databricks

- How do you optimize PySpark jobs for performance in Azure Databricks?
- What are the differences between RDDs and DataFrames in Spark, and when would you use each?
- Explain how Catalyst Optimizer works in Spark SQL.
- How do you handle skewed data in Spark?

Technical: Azure Data Lake Storage Gen2 & Medallion Architecture

- What is the medallion architecture and how have you implemented it in a real-world project?
- How do you manage directory structures and access controls in ADLS Gen2?
- What are best practices for organizing data in Bronze, Silver, and Gold layers?

Technical: Collaboration & Stakeholder Engagement

- How do you gather and translate business requirements into technical specifications?
- Can you share an example of a time you worked closely with analysts or architects to deliver a data solution?

Technical: Data Security & Compliance in Azure

- How do you implement RBAC and ACLs in Azure Data Lake?
- What encryption methods are available in Azure for data at rest and in transit?
- How do you implement data masking in Azure Databricks?

Technical: Data Modeling for Analytics

- What are the differences between star schema, snowflake schema, and data vault modeling?
- How do you approach modeling for real-time vs batch analytics?
- Can you describe a scenario where poor data modeling impacted performance?

Technical: Monitoring & Tuning Databricks Jobs

- What tools do you use to monitor Spark jobs in Databricks?
- How do you troubleshoot job failures or performance bottlenecks?
- What metrics do you track to ensure cost-effective pipeline execution?

Technical: CI/CD & Infrastructure-as-Code

- How do you set up CI/CD pipelines for data engineering workflows using Azure DevOps or GitHub Actions?
- What are the benefits of using Terraform or Bicep for managing Azure resources?
- Can you walk through a deployment pipeline you've built for a data solution?

Technical: Data Quality & Governance

- What data quality checks do you implement in your pipelines?
- How do you enforce data governance policies across your platform?
- What tools or frameworks do you use for data lineage and auditing?

Technical: Mentorship & Best Practices

- How do you mentor junior engineers in your team?
- What coding and design best practices do you promote in your projects?
- How do you ensure code quality and maintainability in collaborative environments?

Technical: Technical Deep-Dive Questions

- Write a PySpark job that reads from ADLS Gen2, performs transformations, and writes to a Gold layer.
- How would you implement schema evolution in a Delta Lake table?
- How do you handle slowly changing dimensions (SCD) in your data pipelines?

Behavioral: Designing & Building Scalable Data Pipelines

- Tell me about a time when you had to design a data pipeline that handled a large volume of data. What challenges did you face, and how did you overcome them?
- Describe a situation where you had to integrate both structured and unstructured data sources. How did you approach this, and what were the results?

Behavioral: Optimizing Spark Jobs in Azure Databricks

- Can you share an example of a Spark job that was running inefficiently? What steps did you take to diagnose and optimize it?
- Describe a time when you had to debug a failing Databricks notebook in production. What tools or techniques did you use to identify the issue?

Behavioral: Implementing Medallion Architecture in ADLS Gen2

- Tell me about a project where you implemented the Bronze-Silver-Gold architecture. How did you decide v
- Have you ever had to restructure an existing data lake to follow best practices? What was your approach?

Behavioral: Stakeholder Collaboration

- Describe a time when business requirements were unclear or frequently changing. How did you manage e
- Tell me about a situation where you had to bridge the gap between technical and non-technical stakeholder

Behavioral: Data Security & Compliance

- Have you ever implemented a security model for a data platform in Azure? What were the key considerations?
- Tell me about a time when you had to ensure compliance with data privacy regulations (e.g., GDPR). How did you approach it?

Behavioral: Data Modeling for Analytics

- Describe a scenario where poor data modeling led to performance issues or incorrect results. How did you
- Tell me about a time when you had to choose between different modeling approaches (e.g., star vs. snowf

Behavioral: Monitoring & Troubleshooting

- Can you share an experience where a data pipeline failed unexpectedly? How did you identify the root cause?
- Describe a time when you had to balance performance and cost in a cloud-based data solution. What trade-offs did you make?

Behavioral: CI/CD & Infrastructure-as-Code

- Tell me about a time when you automated the deployment of a data pipeline. What tools did you use, and what was the outcome?
- Describe a situation where IaC helped you manage infrastructure changes efficiently. How did it improve your workflow?

Behavioral: Data Quality & Governance

- Have you ever implemented a data quality framework across a platform? What checks did you include and
- Tell me about a time when data quality issues impacted business decisions. How did you address the root

Behavioral: Mentorship & Leadership

- Describe a time when you mentored a junior engineer. What approach did you take to help them grow?
- Tell me about a situation where you had to enforce best practices in a team that was resistant to change. How did you handle it?

Tips for Answering Behavioral Questions

- Use the STAR Method: Structure your answers using Situation, Task, Action, Result.
- Tailor Your Examples to the Role: Highlight Azure, Databricks, pipeline design, and collaboration.
- Quantify Your Impact: Include metrics like performance improvements, cost savings, and business impact.
- Show Leadership & Mentorship: Mention mentoring, promoting best practices, and leading initiatives.
- Reflect on Lessons Learned: Be honest about challenges and focus on growth and adaptability.
- Demonstrate Technical Depth: Mention tools, architectural decisions, and trade-offs.
- Practice Clear Communication: Avoid jargon, be concise, and use storytelling.